W10L1

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Countable Sets.

1. A finite set is countable
2. An infinite set is countable if it has the same no. of elements as that of set of natural numbers.

$$\mathbb{N} = \{0, 1, 2, \ldots\}$$

$$A \subseteq \mathbb{N}, \quad A = \{0, 2, 4, 6, \ldots\}$$

$$f : \mathbb{N} \to \underline{\underline{A}}, \qquad f(n) = 2n$$

$$f(0) = \underline{0}, \quad f(1) = \underline{2}, \quad f(2) = \underline{4}, \quad f(3) = \underline{6}, \ldots$$

$$f^{-1}(0) = 0, \quad f^{-1}(2) = 1, \quad f^{-1}(4) = 2, \ldots$$

$\mathbb{Q}$ = set of rational number.

$$\mathbb{Q}^{+} = \{ p/q \mid q \neq 0, \gcd(p,q) = 1, \; p, q \in \mathbb{N} \}$$

is $\mathbb{Q}$ countable.?

Theorem: $\mathbb{Q}$ is countable.

Diagonalization Argument.

$$
\begin{array}{ccccccc}
\frac{1}{1} & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \frac{1}{6} & \cdots \\
\frac{2}{1} & \frac{2}{2} & \frac{2}{3} & \frac{2}{4} & \frac{2}{5} & \frac{2}{6} & \cdots \\
\frac{3}{1} & \frac{3}{2} & \frac{3}{3} & \frac{3}{4} & \frac{3}{5} & \frac{3}{6} & \cdots \\
\frac{4}{4} & \cdots & & & & & \\
\vdots & & & & & &
\end{array}
$$

$\mathbb{Q}$ is contable. $\square$

$\mathbb{R}$ = set of real nubers

is R contable?

Claim: $\mathbb{R}$ is uncoutable.

Proof: Assume $\underline{\mathbb{R}}$ is contable
proof by contradiction

imagine $I = (0, 1)$

$r_0, r_1, r_2, r_3, \ldots$

$r_0 = 0.\underline{1}372095\ldots$

$r_1 = 0.5\underline{2}369173\ldots$

$r_2 = 0.11\underline{2}0201545$

$r_3 = 0.752\underline{5}6481\ldots$

$\vdots$

$x = 0.4334$

" $\sqrt{2}$ is irrational "

Decidable & Recognizable

Theorem: Some languages are not Turing-recognizable.

Outline of the proof

Σ , " Σ^* is countable "

" Set of all languages over Σ is uncountable. "

An undecidable Languge.

$A_{TM} = \{ \underline{\langle M, w \rangle} \mid M$ is a TM that accepts $w \}$

Theorem: "A_{TM} is undecidable."

<u>Proof</u>: Proof by contradiction.

Assume A_{TM} is decidable

Assume there exist a TM H s.t.

H decides A_{TM}.

$$H(\langle M, w \rangle) = \begin{cases} \text{accept} & \text{if } M \text{ accepts } w \\ \text{reject} & \text{if } M \text{ rejects } w \end{cases}$$

Let us create another TM D.

$D =$ " On input $\langle M \rangle$, when M is a TM:

1. Run H on input $\langle M, \langle M \rangle \rangle$

2. Output the opposite of H's output."

$$D(\langle M \rangle) = \begin{cases} \text{accept} & \text{if } M \text{ does not accept } \langle M \\ \text{reject} & \text{if } M \text{ accepts } \langle M \rangle \end{cases}$$

$$D(\langle D \rangle) = \begin{cases} \text{accept} & \text{if } D \text{ does not accept } \langle D \rangle \\ \text{reject} & \text{if } D \text{ accepts } \langle D \rangle \end{cases}$$